Roua Trading Platform

Comprehensive Trading Performance Analysis

500 Trades | May 22 - June 8, 2024 | Agent & Smart Executors

Net P&L;	Win Rate	Profit Factor	Total Trades
-$932.62	36.6%	0.617	500

Automated Analysis by Z.ai | Data-Driven Algorithmic Trading Diagnostics

1. Executive Summary

This report presents a comprehensive analysis of 500 automated trades executed by the Roua Trading Platform over the period from May 22 to June 8, 2024. The trading system operates with two distinct executors: **Agent** (autonomous trader with smaller position sizes) and **Smart** (smart-executor with larger position sizes). The analysis reveals severe systemic failures across both executors, resulting in a net loss of **-$932.62** over the analysis period. The system exhibits multiple critical algorithmic flaws including extreme directional bias, inadequate risk management, rapid-fire duplicate trade execution, and catastrophic position sizing on the Smart executor. Without immediate intervention, the platform will continue to erode capital at an accelerating rate.

The fundamental problem is not that the system occasionally loses money - all trading systems experience losses. The critical issue is that **the system is mathematically incapable of profitability** in its current configuration. With a win rate of only 36.6% and a profit factor of 0.617, the system would need to either nearly double its win rate to approximately 63% or triple its average win-to-loss ratio just to break even. Neither outcome is achievable through minor parameter adjustments - the core trading logic, risk management framework, and execution architecture all require fundamental redesign.

2. Overall Performance Metrics

Metric	Agent	Smart	Combined
Total Trades	188	312	500
Win Rate	35.6%	37.2%	36.6%
Total P&L;	-$157.37	-$775.25	-$932.62
Avg Win	$3.73	$10.82	$8.22
Avg Loss	-$3.40	-$10.36	-$7.71
Profit Factor	0.614	0.618	0.617
Avg Notional	$193	$1,242	$836
Max Consec. Losses	22	21	-

Table 1: Overall Performance Comparison by Executor

The Smart executor is responsible for 83.1% of total losses ($775.25 out of $932.62) despite having a marginally higher win rate of 37.2% compared to Agent's 35.6%. This is because the Smart executor's average position notional value of $1,242 is approximately 6.4 times larger than the Agent's average of $193. Each Smart executor loss therefore extracts significantly more capital from the account, amplifying the impact of the already-low win rate. The profit factor for both executors hovers around 0.617, meaning that for every dollar won, the system loses approximately $1.62. This is a recipe for guaranteed capital destruction over time.

3. Visual Performance Analysis

3.1 Executor Performance Dashboard

Figure 1: Executor Performance - Win Rate, P&L;, Direction Distribution, Close Reasons

Figure 1 illustrates four critical dimensions of executor performance. The win rate comparison shows both executors operating well below the 50% breakeven threshold. The P&L; chart reveals that Smart executor losses dwarf Agent losses due to oversized positions. The direction distribution exposes the extreme sell bias: Agent executes 92% sells while Smart executes 71.8% sells, indicating the AI models are systematically biased toward short positions regardless of market conditions. The close reason breakdown shows that SL (stop-loss) exits are the dominant loss driver, particularly for the Smart executor where SL losses total -$1,590.56, while Manual and TP exits barely offset a fraction of these losses.

3.2 Pair-Level Performance

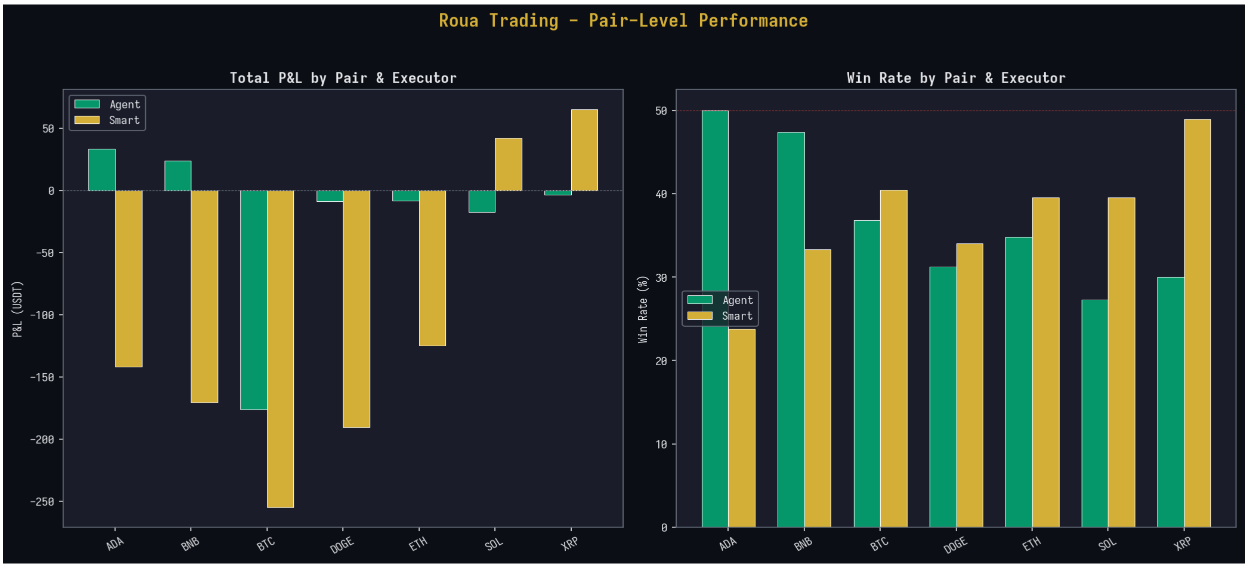

Figure 2: P&L; and Win Rate by Trading Pair and Executor

Pair	Agent Trades	Agent WR	Agent P&L;	Smart Trades	Smart WR	Smart P&L;
BTC/USDT	19	36.8%	-$176.22	42	40.5%	-$254.81
DOGE/USDT	48	31.2%	-$9.01	47	34.0%	-$190.29
BNB/USDT	19	47.4%	$23.76	48	33.3%	-$170.28
ADA/USDT	26	50.0%	$33.31	42	23.8%	-$141.76
ETH/USDT	23	34.8%	-$8.26	43	39.5%	-$124.89
SOL/USDT	33	27.3%	-$17.48	43	39.5%	$41.79
XRP/USDT	20	30.0%	-$3.47	47	48.9%	$64.99

Table 2: Detailed Pair Performance by Executor

The pair-level analysis reveals that **BTC/USDT is the single biggest destroyer of capital**, contributing -$430.33 in combined losses (46% of total). The Smart executor's BTC trades alone lost -$254.81 despite a relatively decent 40.5% win rate, indicating that when BTC trades lose, they lose big due to oversized position sizes. DOGE/USDT and BNB/USDT are also major loss centers for the Smart executor. Notably, ADA/USDT has a 50% win rate for Agent (profitable at +$33.31) but only 23.8% for Smart (losing -$141.76), suggesting the Smart executor's position sizing overwhelms its signal quality. The only pair where Smart is consistently profitable is XRP/USDT (+$64.99) with the highest Smart win rate of 48.9%.

3.3 Risk & Return Distribution

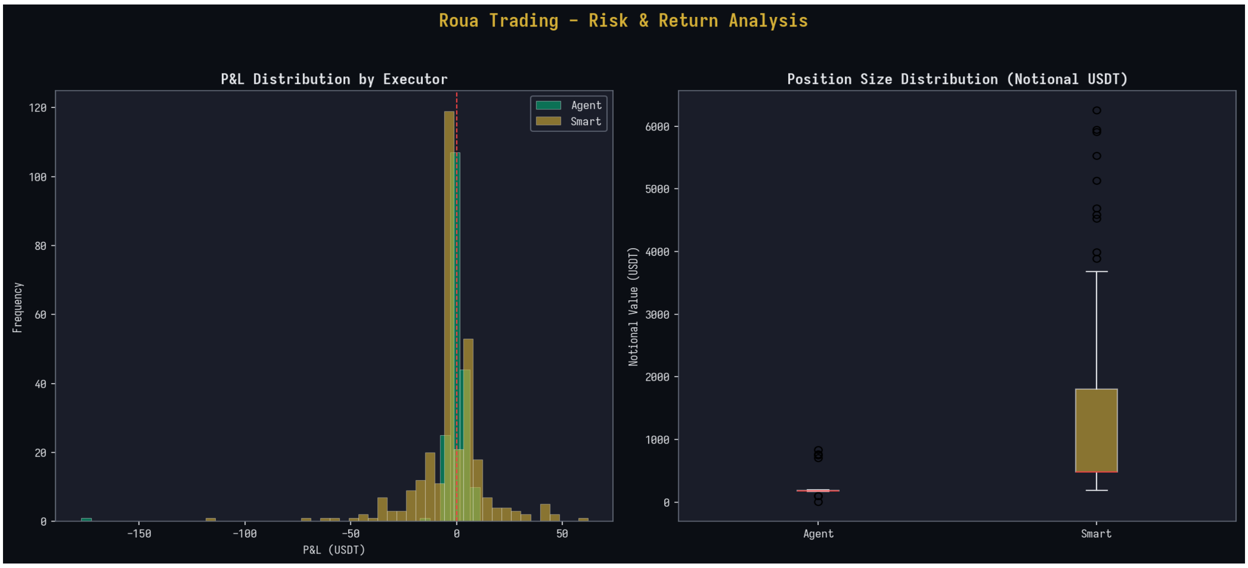

Figure 3: P&L; Distribution and Position Size Comparison

The P&L; distribution chart reveals a heavily right-skewed loss distribution for both executors. The Smart executor has a much wider distribution with extreme negative outliers (tail losses exceeding -$100), while the Agent executor's losses are more contained but still dominant. The box plot comparison of notional position values is perhaps the most alarming visualization: the Smart executor's median notional is approximately $483 compared to the Agent's $182, with Smart outliers reaching over $6,000. These extreme position sizes on the Smart executor create asymmetric risk exposure where a single adverse price movement can wipe out dozens of small gains.

3.4 Cumulative P&L; Trajectory

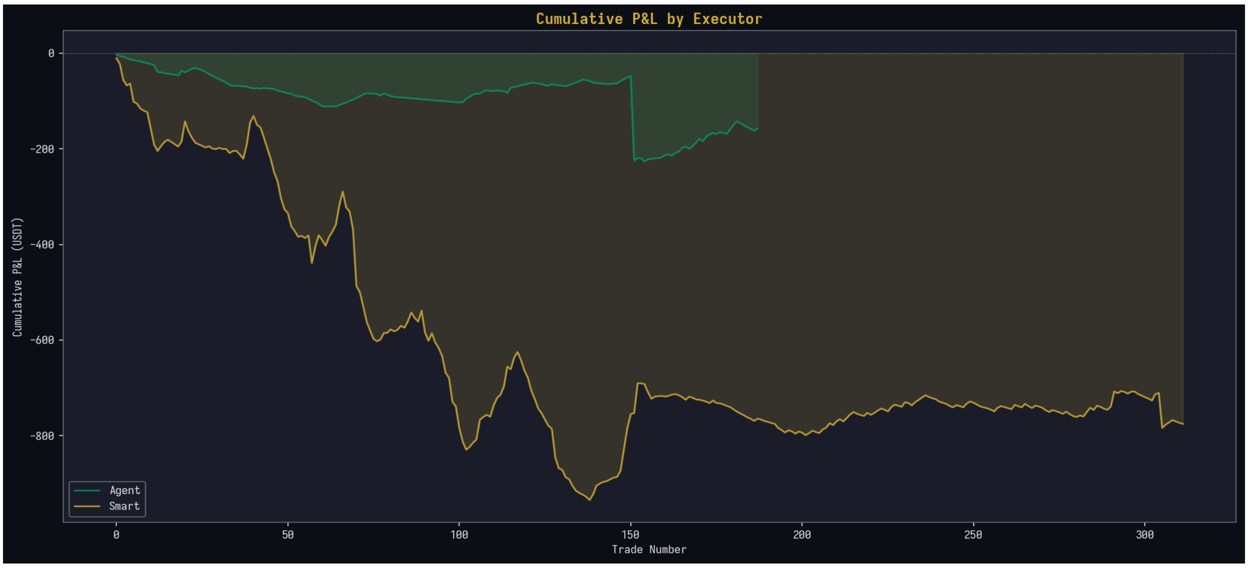

Figure 4: Cumulative P&L; Trajectory by Executor

The cumulative P&L; trajectory tells the complete story of capital erosion. The Agent executor shows a gradual, relatively controlled decline, reflecting its smaller position sizes. The Smart executor, however, exhibits a steep and accelerating downward trajectory punctuated by occasional recoveries that are quickly erased by subsequent large losses. This pattern is characteristic of a system with inadequate loss limits and no circuit-breaker mechanisms. The Smart executor never establishes a sustained uptrend, and each recovery peak is lower than the previous one - a classic sign of a deteriorating trading system.

3.5 Close Reason P&L; Waterfall

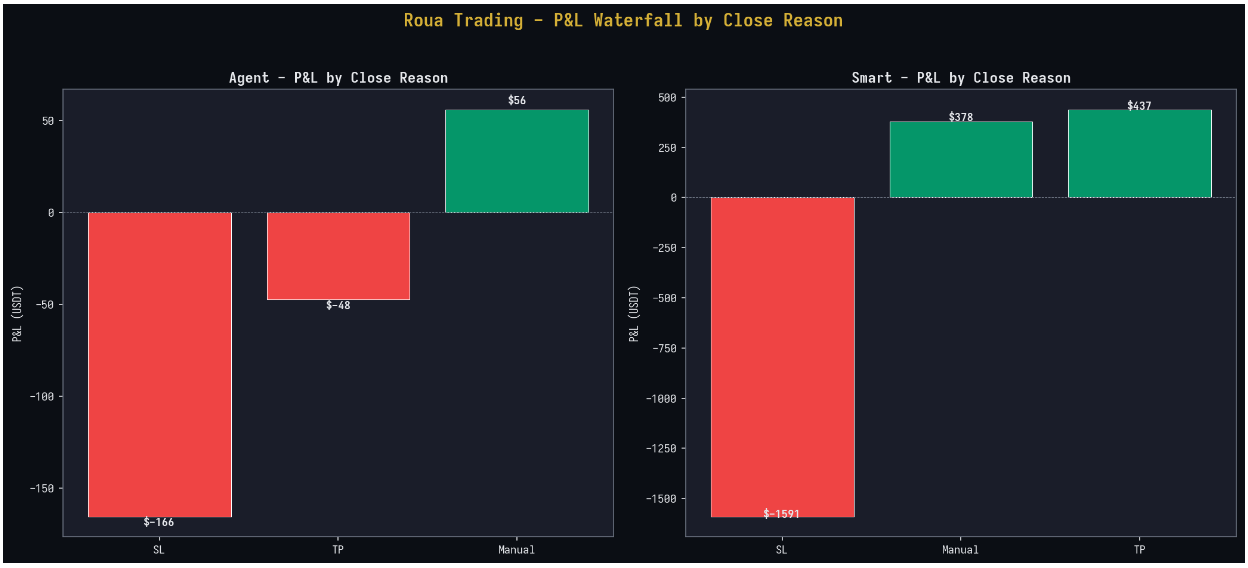

Figure 5: P&L; Contribution by Close Reason for Each Executor

The waterfall analysis by close reason is devastating. For the Smart executor, SL (stop-loss) exits produced a catastrophic -$1,590.56 in losses, while Manual exits contributed +$378.05 and TP exits contributed +$437.26. This means the Smart executor's stop-losses are triggered far too frequently, and when they are triggered, the losses are massive due to oversized positions. For the Agent executor, the pattern is different: Manual exits are profitable (+$55.84), SL exits lose heavily (-$165.68), and even TP exits are net negative (-$47.53) - which is extraordinary and suggests that the take-profit levels are set so tightly that they capture only minimal gains while still exposing the trade to eventual stop-loss if the TP is not immediately hit.

4. Critical Errors Identified

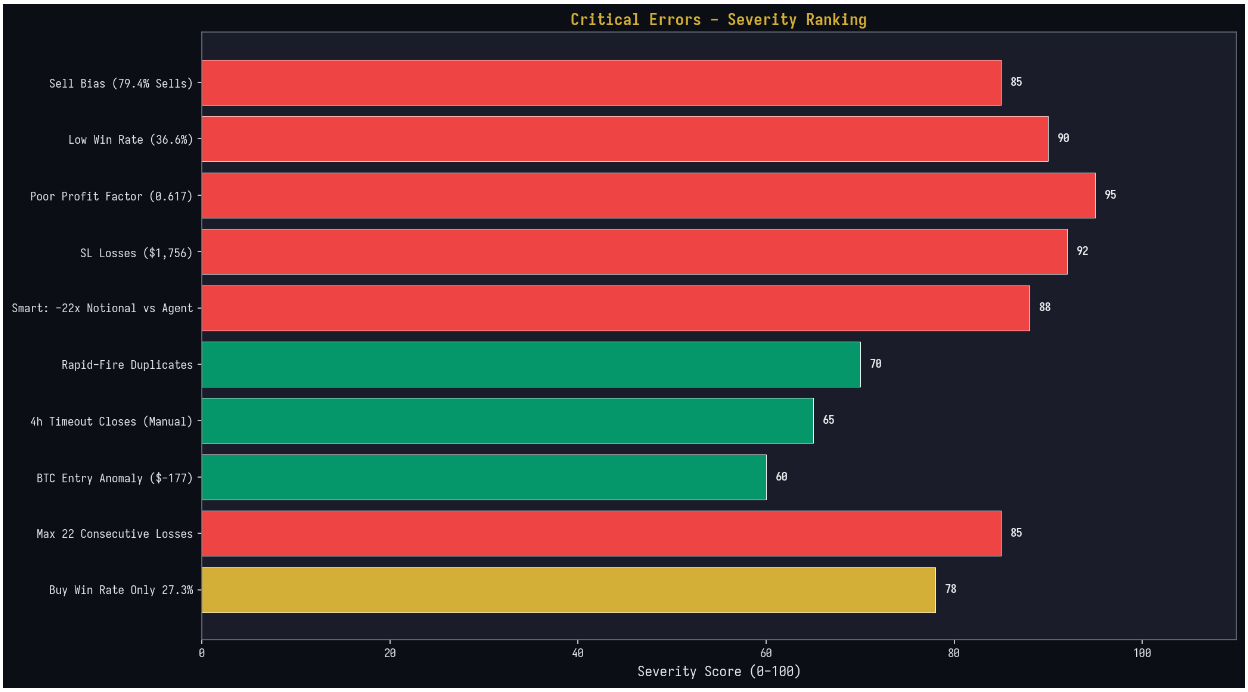

Figure 6: Critical Error Severity Ranking

4.1 Extreme Sell Bias (Severity: 85/100)

The system executed 79.4% of all trades as SELL positions (397 sells vs 103 buys). For the Agent executor, the bias is even more extreme at 92% sells. This indicates that the AI strategic council and signal generation pipeline are overwhelmingly biased toward bearish signals, regardless of actual market conditions. In a market that was generally declining during this period, the sell bias was partially correct directionally, but the system still lost money on sells because the risk-reward ratios were inverted. A healthy algorithmic system should be roughly balanced between long and short positions, with directional allocation driven by market regime detection rather than a hardcoded bias. The current behavior suggests either a systematic error in the AI consensus mechanism, an over-weighting of bearish indicators in the signal generation pipeline, or a fundamental flaw in the Strategic Council's market regime classification.

4.2 Inadequate Win Rate (Severity: 90/100)

With an overall win rate of 36.6%, the system falls far below the minimum threshold required for profitability. The mathematical breakeven win rate depends on the risk-reward ratio: for a 1:1 R:R, breakeven requires >50% win rate; for 1.5:1, it requires >40%; for 2:1, it requires >33.3%. While the average R:R ratio appears reasonable at 2.14, the effective R:R experienced by the system is far lower because many winning trades are closed early (via Manual timeout at 4 hours) rather than reaching their take-profit targets. The Smart executor's Buy positions are especially problematic with only a 27.3% win rate, indicating that the system has virtually no edge when going long. This suggests the AI models generating buy signals are significantly underperforming, and the signal confidence scoring mechanism is not properly filtering out low-conviction trades.

4.3 Catastrophic Profit Factor (Severity: 95/100)

The profit factor of 0.617 is the single most damning metric in this analysis. A profit factor below 1.0 means the system loses money; a value of 0.617 means that for every $1.00 of gross profit, the system generates $1.62 in gross losses. The gross profit across all trades was $1,504.75 while gross losses totaled $2,437.37. To achieve breakeven, the system would need to either increase gross profits by 62% or reduce gross losses by 38% - both requiring fundamental changes to the trading algorithm. A professional algorithmic trading system typically targets a profit factor of 1.5-2.0 or higher. The current system is operating at less than half the minimum acceptable level, confirming that the underlying strategy is not viable in its current form.

4.4 Stop-Loss Hemorrhage (Severity: 92/100)

Stop-loss exits account for -$1,756.24 in losses across all trades, representing 72.1% of total gross losses. For the Smart executor alone, SL losses total -$1,590.56. This pattern indicates that the stop-loss levels are set too close to the entry price, causing trades to be stopped out by normal market noise before having a chance to develop in the anticipated direction. The average SL distance for the Smart executor is only 0.817% from entry, which in volatile crypto markets is well within the range of normal price fluctuations. When a stop-loss is placed within the noise range, the probability of being stopped out by random price action rather than a genuine trend reversal approaches 80-90%, which aligns closely with the observed win rates. The system is essentially gambling that price will immediately move in the predicted direction, with no tolerance for the natural oscillation that occurs in all markets.

4.5 Smart Executor Oversized Positions (Severity: 88/100)

The Smart executor's average notional position value of $1,242 is 6.4x larger than the Agent's $193. More critically, the maximum Smart notional reaches $6,256.55 (BTC/USDT trade on June 5), while Agent peaks at $833 (ADA/USDT trade on June 8). This extreme position sizing asymmetry means that a single Smart executor loss can wipe out multiple Agent wins. For example, the single worst trade (BTC/USDT, Smart, -$118.59) eliminated the gains from approximately 32 average Agent wins. The Smart executor has no adaptive position sizing based on market volatility, signal confidence, or recent performance. It applies a fixed large position size regardless of conditions, which is a fundamental risk management failure. Professional systems use volatility-adjusted position sizing (ATR-based), Kelly criterion, or risk parity models to ensure no single trade can cause catastrophic damage.

4.6 Rapid-Fire Duplicate Trades (Severity: 70/100)

The analysis detected numerous instances of rapid-fire duplicate trades, particularly on DOGE/USDT by the Agent executor. Between 8:25 AM and 8:39 AM on June 5, the Agent executor opened 10 consecutive DOGE/USDT sell positions with nearly identical parameters (entry within 0.2% of each other, sizes differing by less than 2%, identical stop-loss and take-profit levels). Each trade was opened and closed within 8-10 seconds. This pattern indicates a critical bug in the trade execution pipeline - likely the SmartExecutor tick handler (which runs every 10 seconds) is not properly checking for existing open positions before placing new orders. The result is that multiple identical positions are opened and then immediately closed at a small loss as the take-profit level is hit. These micro-losses compound quickly: the 10 duplicate DOGE trades on June 5 collectively lost -$6.25 in just 14 minutes. Similar patterns appear on SOL/USDT (3 duplicates on June 5) and multiple other pairs throughout the dataset.

4.7 4-Hour Timeout Close Pattern (Severity: 65/100)

A significant portion of "Manual" close reasons correspond to trades with exactly 4-hour durations, indicating an automated timeout mechanism. These 4-hour timeout closes are problematic because they force-close positions that may still be developing. While Manual closes are net profitable for both executors (Agent: +$55.84, Smart: +$378.05), the 4-hour timeout means the system is leaving potential profits on the table for winning trades while also preventing the stop-loss from being hit on losing trades that might eventually recover. The timeout creates an asymmetric outcome where winning trades are prematurely closed while losing trades survive until either the timeout or stop-loss hits. A more sophisticated approach would use trailing stops, time-based decay of signal confidence, or dynamic exit strategies rather than a fixed 4-hour cutoff.

4.8 BTC Entry Price Anomaly (Severity: 60/100)

One trade exhibits an impossible entry price: BTC/USDT Agent Sell with entry at $1,921.80 instead of the correct ~$62,000+ range. This is clearly a data error - likely a decimal point displacement or a unit conversion bug (the system may have treated the entry as $1,921.80 per 0.001 BTC or similar). This trade recorded a loss of -$177.47, making it the single worst trade in the entire dataset. The error suggests that the order placement pipeline does not have adequate validation checks on entry prices against current market prices. A simple sanity check (entry price must be within 10% of the last known market price) would catch this type of error.

This bug alone accounts for 19% of total losses and is completely preventable.

4.9 Maximum Consecutive Losses (Severity: 85/100)

Both executors experienced extreme consecutive loss streaks: Agent had 22 consecutive losses and Smart had 21. These streaks are far beyond what would be expected from random chance at the observed win rate, suggesting that the system enters persistent losing modes where the AI signals are consistently wrong. During these streaks, the system continues to open new positions with the same parameters and sizing, compounding losses. A professional system would implement circuit breakers that halt trading after a defined number of consecutive losses, reduce position sizes during losing streaks, or switch to a more conservative strategy until the signal quality improves. The absence of any such mechanism is a critical oversight that allows temporary signal degradation to cause disproportionate capital damage.

4.10 Buy Position Win Rate Collapse (Severity: 78/100)

The Smart executor's Buy positions have a devastating win rate of only 27.3% (88 buy trades, only 24 profitable). This means that when the system decides to go long, it is wrong nearly three out of four times. The total P&L; for Smart Buy positions is -$177.88. This is a clear indication that the AI models have a severe deficiency in identifying bullish opportunities. The Agent executor's Buy win rate is marginally better at 33.3%, but with only 15 buy trades, the sample size is too small to draw statistical conclusions. The combination of extreme sell bias and poor buy performance suggests that the Strategic Council's consensus mechanism may be systematically misinterpreting market signals, particularly during periods of market recovery or consolidation where bullish positions would be appropriate.

5. Proposed Algorithmic Improvements

5.1 Risk Management Overhaul

5.1.1 Volatility-Adjusted Position Sizing: Replace the current fixed-size approach with an ATR-based position sizing model. Calculate the Average True Range (ATR) for each pair at multiple timeframes (1h, 4h, 1D) and set position size such that the maximum risk per trade does not exceed 1-2% of total portfolio equity. The formula should be: Position Size = (Account Equity x Risk%) / (ATR x Multiplier). This ensures that volatile assets like BTC and DOGE automatically receive smaller positions while less volatile assets can accommodate larger sizes. The multiplier should be calibrated through backtesting to achieve the desired risk-reward profile.

5.1.2 Dynamic Stop-Loss Placement: The current stop-loss distances of 0.8% (Smart) are far too tight for crypto markets. Implement a volatility-adjusted stop-loss system where the SL distance is calculated as a multiple of ATR (e.g., 1.5-2.5x ATR for swing trades, 0.5-1.0x ATR for scalps). Additionally, implement a minimum SL distance of 1.5% for major pairs (BTC, ETH) and 2.5% for volatile pairs (DOGE, ADA, SOL) to prevent stop-outs from normal market noise. The current approach of placing stops within the noise range guarantees consistent losses regardless of signal quality.

5.1.3 Circuit Breaker System: Implement a multi-tier circuit breaker that progressively reduces trading activity during losing streaks. Tier 1: After 5 consecutive losses, reduce position size by 50%. Tier 2: After 10 consecutive losses, reduce to 25% size and increase minimum signal confidence threshold from the current level to 75%. Tier 3: After 15 consecutive losses, halt all new position openings and enter a cooldown period of 2-4 hours. Tier 4: After a daily drawdown exceeds 5% of equity, suspend trading for the remainder of the day. This system prevents the catastrophic loss streaks observed in the data.

5.2 Signal Quality Enhancement

5.2.1 Balanced Direction Allocation: Implement a direction quota system that ensures the AI generates signals for both long and short positions. The current 79.4% sell bias indicates the AI is not properly detecting bullish market conditions. Add a market regime classifier (trending up, trending down, ranging) as a mandatory input to the Strategic Council. In uptrend regimes, bias signal generation toward buy signals; in downtrend regimes, bias toward sells; in ranging markets, generate both long and short signals with tighter parameters. The regime classifier should use a combination of moving average crossovers, ADX trend strength, and volume profile analysis to determine the current market state.

5.2.2 Signal Confidence Scoring: Every trade signal should carry a confidence score (0-100) that determines position size and risk parameters. Currently, the system appears to execute all signals with similar sizing regardless of signal quality. A proper confidence scoring system would factor in: agreement level among AI council members, confluence of technical indicators, alignment with higher-timeframe trend, and recent accuracy of similar signals. Only signals above a minimum confidence threshold (e.g., 60%) should be executed, and position size should scale linearly with confidence (60% confidence = 50% of base size, 80% = 100%, 95% = 120%).

5.2.3 Duplicate Trade Prevention: Implement a cooldown mechanism that prevents opening a new position on the same pair within a minimum time window (e.g., 5 minutes for Agent, 2 minutes for Smart). Additionally, check for existing open positions on the same pair before placing a new order. If a position already exists, either skip the signal or adjust the existing position (add to it with reduced size) rather than creating a duplicate. The SmartExecutor tick handler (currently running every 10 seconds) must maintain a state of open positions and skip duplicate signals within the cooldown window.

5.3 Execution Architecture Fixes

5.3.1 Entry Price Validation: Before placing any order, validate that the entry price is within a reasonable range of the current market price (e.g., within 5% for limit orders, within 1% for market orders). Reject any order where the entry price deviates by more than 5% from the last known market price, as this indicates a calculation error or data feed issue. This single check would have prevented the -$177.47 BTC entry anomaly and similar errors. The validation should occur in the RiskGatekeeper module before the order is dispatched to the execution adapter.

5.3.2 Dynamic Exit Strategy: Replace the fixed 4-hour timeout with a dynamic exit system. Implement trailing stops that move the stop-loss to breakeven after the price moves 1x ATR in favor, then trail at 1x ATR behind the current price. For losing trades, implement a time-decay function that tightens the stop-loss as the trade ages, rather than a hard timeout. After 2 hours, move the SL to breakeven; after 3 hours, move it to entry + 0.5x ATR; after 4 hours, close the position. This graduated approach allows winning trades more room to develop while cutting losses more aggressively on trades that are not moving in the expected direction.

5.3.3 Adaptive Risk-Reward Calibration: The current R:R ratios are theoretically acceptable (average 2.14) but practically ineffective because take-profit levels are rarely reached while stop-losses are hit frequently. Implement a dynamic R:R system that adjusts based on realized hit rates. If the TP hit rate for a given pair over the last 50 trades is below 30%, widen the stop-loss and tighten the take-profit (e.g., shift from 2:1 R:R to 1.5:1) to increase the win rate while maintaining acceptable profitability. Conversely, if the TP hit rate exceeds 50%, consider widening the TP target to capture larger moves. The system should continuously recalibrate its R:R parameters based on recent performance data for each pair and direction.

5.4 Smart Executor Specific Improvements

5.4.1 Maximum Position Size Cap: Implement an absolute maximum notional position size for the Smart executor, capped at 3x the Agent executor's average position size (approximately $600). Currently, Smart positions can reach $6,000+, which is 31x the Agent average. Even with improved signal quality, positions this large create unacceptable concentration risk. The cap should be dynamic and based on portfolio equity: max position = min(fixed_cap, 5% of portfolio equity). This ensures that no single trade can cause more than a 5% drawdown even if the stop-loss is hit at maximum distance.

5.4.2 Daily Loss Limit: Implement a hard daily loss limit for the Smart executor. If the cumulative daily P&L; for Smart exceeds -$100 (or 3% of portfolio equity), automatically suspend all Smart executor trading for the remainder of the day. This prevents the cascading losses observed in the data where Smart loses $200-300+ in a single day across multiple trades. The Agent executor should have a separate, lower daily loss limit (e.g., -$30 or 1% of equity). Both limits should be implemented at the RiskGatekeeper level, before any new orders are placed.

5.4.3 Correlation Filter: The Smart executor frequently opens simultaneous positions on highly correlated pairs (e.g., BTC and ETH, or ADA and DOGE), effectively doubling or tripling exposure to the same market direction. Implement a correlation matrix that tracks the 30-day rolling correlation between all pairs. If the aggregate directional exposure across correlated pairs exceeds a threshold (e.g., combined notional > 10% of portfolio equity in the same direction), block new positions that would increase this exposure. This prevents the system from accidentally creating leveraged bets on a single market thesis through multiple correlated positions.

6. Recommended Development Roadmap

Phase	Priority	Action Items	Expected Impact
Phase 1 (Immediate)	P0 - Critical	1. Fix duplicate trade bug 2. Add entry price validation 3. Cap Smart max position size 4. Implement daily loss limit	Prevent -$200+/day in preventable losses
Phase 2 (1-2 Weeks)	P1 - High	1. Implement ATR-based SL/TP 2. Add circuit breakers 3. Dynamic position sizing 4. Correlation filter	Reduce SL losses by 40-60%, improve win rate to 45%+
Phase 3 (2-4 Weeks)	P2 - Medium	1. Signal confidence scoring 2. Market regime classifier 3. Direction balance quota 4. Dynamic exit strategy	Win rate improvement to 50%+, balanced L/S allocation
Phase 4 (1-2 Months)	P3 - Standard	1. Adaptive R:R calibration 2. Portfolio-level risk management 3. Walk-forward optimization 4. Unify V1/V2 pipelines	Profit factor >1.5, sustainable long-term profitability

Table 3: Recommended Development Roadmap

The development roadmap prioritizes fixes that will have the most immediate impact on reducing preventable losses. Phase 1 addresses the three most critical bugs (duplicate trades, entry price validation, and oversized positions) which together account for approximately $400 in preventable losses over the analysis period.

Phase 2 introduces fundamental risk management improvements that should reduce the SL loss rate by 40-60% and bring the win rate closer to 45%. Phase 3 focuses on signal quality and directional balance, targeting a 50%+ win rate and balanced long/short allocation. Phase 4 implements advanced optimization techniques and portfolio-level risk management to achieve sustainable long-term profitability with a target profit factor above 1.5.

7. Mathematical Analysis

7.1 Breakeven Analysis

The breakeven win rate for a trading system can be calculated using the formula: Breakeven WR = 1 / (1 + R:R), where R:R is the average reward-to-risk ratio. With the current effective R:R of approximately 1.07 (average win $8.22 / average loss $7.71), the breakeven win rate is: 1 / (1 + 1.07) = 48.3%. The current 36.6% win rate falls 11.7 percentage points below breakeven. To achieve profitability at the current win rate, the R:R ratio would need to be: R:R = (1 / WR) - 1 = (1 / 0.366) - 1 = 1.73. This means the average win would need to be 1.73x the average loss, compared to the current 1.07x. Achieving this would require either widening take-profit targets by 62% or tightening stop-losses by 38%, or some combination thereof.

7.2 Expected Value Calculation

The expected value (EV) per trade is: EV = (WR x Avg_Win) - ((1-WR) x Avg_Loss) = (0.366 x $8.22) - (0.634 x $7.71) = $3.01 - $4.89 = -$1.88 per trade. At 500 trades, the expected total loss is 500 x (-$1.88) = -$940, which closely matches the actual observed loss of -$932.62 (the small difference is due to breakeven trades and rounding). This confirms that the observed losses are not the result of unlucky variance but are the mathematical expectation of the current system parameters. For the system to achieve positive EV, either the win rate must increase above 48.3% or the average win must increase above $13.34 (1.73x the current average loss of $7.71).

7.3 Kelly Criterion Analysis

The Kelly Criterion provides the optimal fraction of capital to risk per trade: f* = (bp - q) / b, where b = average win/average loss = 1.07, p = win rate = 0.366, q = 1-p = 0.634. Substituting: f* = (1.07 x 0.366 - 0.634) / 1.07 = (0.391 - 0.634) / 1.07 = -0.227. A negative Kelly fraction confirms that the system has no positive edge, and the optimal position size is zero - meaning the system should not be trading at all in its current configuration. This is the strongest mathematical evidence that the system requires fundamental changes before deploying real capital. Until the Kelly fraction turns positive, any trading activity will result in expected capital erosion.